Trifecta Tech Foundation

2025
annual report

Building blocks that make the digital world safer for everyone

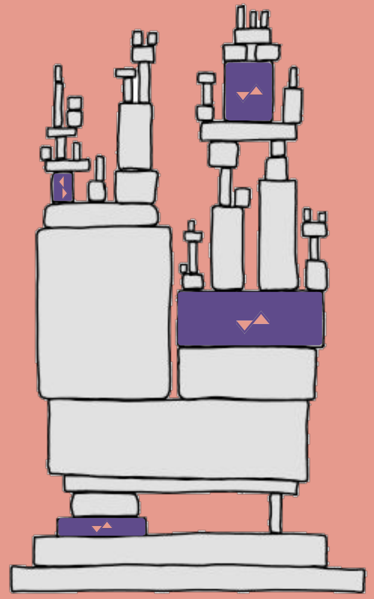
open source - independent - non-profit

About

Trifecta Tech Foundation is a public benefit organization founded in April of 2024 as the long-term guardian of critical pieces of open source software: *building blocks* that are used in the infrastructure layer of the Internet and vital systems; invisible pieces of software we all rely on.

Our mission is to impact the digital security of hundreds of millions of people by decreasing the attack surface of those systems. Our engineers create and maintain robust and inherently safer building blocks, and make them available to everyone to use freely.

We are committed to open source and open standards as the single best way to create the most impact for the public. Where we can we support the open source communities we rely on, in particular the Rust programming language: it provides better safety and security guarantees, allows us to *innovate without breaking things,* and - as a result - maximizes our impact.



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2025 in brief

2025 was our first year of full operation, and we're proud to say it was incredibly successful. Sudo-rs was adopted by Ubuntu, ntpd-rs continued to run like a charm at Let's Encrypt, and our data compression libraries zlib and bzip reached millions of users, with many more to come.

We started work on a new and ambitious project, "Secure time for a Safe Internet", which aims to provide reliable time to everyone on the Internet. The goal is to dramatically speed up adoption of NTS (Network Time Security), similar to the adoption of HTTPS in the late 2010s.

We hired two staff members, as the first step towards building a strong team of maintainers of open source critical infrastructure.

Our numbers were good; we hit our fundraising targets and - ahead of schedule - started building maintenance reserves; we are all set for 2026 and plan to create even more impact for a safer digital world!

Erik Jonkers - Chair of the Board

Our board

Erik Jonkers
Chair

Hugo van de Pol
Secretary

Marlon Baeten
Treasurer

Our initiatives

Time synchronization
Reliable, securely synchronized time is a building block for the Internet and vital infrastructure.

Privilege boundary
The sudo utility mediates a critical privilege boundary on every open source OS that powers the Internet.

Data compression
Almost all content sent over the Internet undergoes data compression using algorithms like zlib and zstd.

Initiative report

Time Synchronization

trifectatech.org/initiatives/time-synchronization

Sponsors and funders

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Secure time for a Safe Internet

Reliable, securely synchronized time is a requirement for the Internet and other vital infrastructure. No correct time? No TLS certificates, no telecommunications, no energy grid.

Our *Project Pendulum* provides modern implementations of the Network Time Protocol (NTP) and Precision Time Protocol (PTP). Our NTP implementation, ntpd-rs, is packaged for Fedora, Debian and Ubuntu, among others, and deployed at Let's Encrypt.

In 2025, we extended the feature set of ntpd-rs adding support for GNSS time sources and the next version of NTP, version 5 [ntpv5].

We started an ambitious project, *Secure time for a Safe Internet*, that aims to provide reliable time to everyone by speeding up the adoption of NTS (Network Time Security), similar to HTTPS in the late 2010s. We created an experimental service, a pool of NTS servers [pools], and proposed a standard to The Internet Engineering Task Force (IETF). The work will continue in 2026 and early 2027.

Other activities included contributions to the IETF Network Time Protocols working group and attending the IETF 123 meeting in Madrid, Spain.

Memory-safe data compression

Almost all content sent over the Internet undergoes data compression using algorithms like zlib and zstd. Implemented in C, current libraries encounter regular security issues despite receiving extensive industry-wide scrutiny.

Our *Data Compression* initiative creates alternative, memory-safe implementations of these widely-used compression libraries.

In 2025, we brought memory-safe data compression to millions of users via our zlib-rs and bzip2-rs projects. Our activities included researching performance optimizations and running security audits. Zlib-rs reached 30M downloads by developers [30M] and was used in 5.000 open source projects [5k].

We started a new project for the Zstandard algorithm. Zstandard is a modern successor to zlib, providing better compression faster. Similarly to zlib-rs and bzip2-rs we aim to provide excellent performance while introducing memory safety. At the end of 2025, we completed initial development work for the decompression part of our Zstandard software. We expect to release this part in 2026 and start - pending funding - development work on compression.

Initiative report

Data Compression

trifectatech.org/initiatives/data-compression

Sponsors and funders

Ministerie van Binnenlandse Zaken en Koninkrijksrelaties

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Chainguard

ASTRAL

Security is a paramount concern for Firefox. A performance- competitive, memory-safe implementation of zlib and zstd is a very welcome addition to Firefox and will make our users safer.

Bobby Holley – CTO Firefox @ Mozilla

Initiative report

Privilege Boundary

trifectatech.org/initiatives/privilege-boundary

Sponsors and funders

Canonical

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ISRG Internet Security Research Group

Sovereign Tech Agency

Sudo-rs lands in Ubuntu 25.10

During 2025 the development of sudo-rs focussed on portability and adding features for mainstream adoption by Linux distributions[distro], culminating in the release of Ubuntu 25.10 with sudo-rs as the default sudo.

In October, Canonical announced[2510] the release of Ubuntu 25.10: *"the introduction of Trifecta Tech Foundation's memory-safe sudo delivers many benefits compared to the traditional implementation. For users, a reduced attack surface in the sudo tool will improve Ubuntu's overall security posture"*.

The sudo-rs project has come a long way since initial development was started in 2023 by the Internet Security Research Group, as part of the Prossimo project[ISRG]. We're thankful to ISRG and our funders, for making this project possible.

Beyond sudo-rs, our activities in 2025 included research and initial development of two potential new projects in the Privilege Boundary initiative: *allowd* and an SSH (Secure Shell) server.

For 2026, our goals include increased adoption by distributions big and small, and supporting and growing our community of contributors.

Ecosystem activities

We're engineers, and we make change for the public by creating and publishing software. That said, a number of our other activities in 2025 are equally important to our mission:

Standardization: we attended meetings of the IEEE and IETF timing working groups and contributed to evolving standards for NTP and PTP.

Policy: we raised awareness for *Secure by Design* software development practices and published a multi-stakeholder statement "Calling for memory safety incentives in EU cybersecurity policies" [eu-call].

Ecosystem contributions: we contributed to the Rust programming language that we use for our projects, in particular we researched improvements for performance-critical applications [perf].

Education: we added new content to teach-rs, a modular university course teaching the Rust programming language [teach].

We believe that software must become safer; that this is paramount in critical infrastructure; and that Rust is an outstanding choice of technology for this purpose.

Trifecta Tech Foundation's memberships

OpenSSF OPEN SOURCE SECURITY FOUNDATION	THE LINUX FOUNDATION	Rust Foundation
Nonprofit Cyber	OpenInfra FOUNDATION ASSOCIATE MEMBER	

Certifications

Algemeen Nut Beogende Instelling ANBI	NGOsource Equivalency Determination on File

Financials

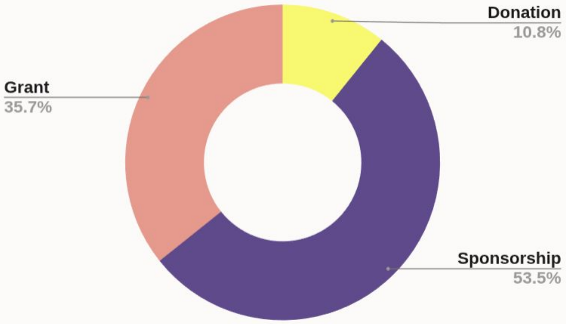
The objective for our fundraising activities is to diversify the income sources: subsidies, donations, sponsorships, and grants should all contribute to our income, and keep the dependence on a single source low.

In 2025, we hit our targets and kept our costs low; lower than budgeted. As a result, we were able to start building our reserves, ahead of schedule.

The income sources in 2025 turned out to be reasonably diverse, originating from the ICANN Grant Program, donations from NLnet Foundation, and sponsorships from companies like Canonical, AWS, Chainguard and Astral.

Our board operated on a voluntary basis and did not receive any compensation in 2025.

Income by source



Income & expenses

The total income for 2025 was €488k, slightly over our prognosis of €475k. Costs, budgeted at €475k, came to a total of €387k, giving us a positive sum of €101k, which was added to our general reserves.

Income (€ 1000)			
	2024 actual	2025 actual	2025 budget
Initiative donations & sponsorships	53	455	425
General donations & sponsorships		29	50
Consultancy income		4	
Research income			
Total	**53**	**488**	**475**

Expenditure (€ 1000)			
	2024 actual	2025 actual	2025 budget
Staff			
Staff (R&D)		60	60
Suppliers (R&D)	43	303	370
Travel		10	18
Housing		2	12
Other costs	6	12	15
Total	**49**	**387**	**475**
Sum of Income and Expendidure	**4**	**101**	**0**

Actual General Reserve	**4**	**105**	

Budget 2026-2027

For next year and for 2027 we plan for a controlled increase of our budget, but a significant growth of our team. We plan to employ at least two more maintainers, the exact number depending on part-time or full-time employment.

In both years, we don't plan for an increase of our general reserve, but in case our numbers are positive we will increase our reserve.

Expected income (€ 1000)		
	2026	**2027**
Initiative donations & sponsorships	410	425
General donations & sponsorships	80	200
Consultancy income	20	20
Research income	25	25
Total	**535**	**670**

Expected expenditure (€ 1000)		
	2026	**2027**
Staff	30	32
Staff (R&D)	250	325
Suppliers (R&D)	175	190
Suppliers (Other)	20	25
Travel	20	30
Housing	10	18
Other costs	30	50
Total	**535**	**670**
Sum of Income and Expenditure	**0**	**0**

General Reserve	**105**	**105**

Looking ahead

We have activities and projects planned for our initiatives *Time synchronization*, *Privilege boundary* and *Data compression*, which we expect will increase the adoption, and therefore the impact of our work. Completing those activities successfully will be our main goal.

In our new projects, we will invest significantly in *Secure time for a Safe Internet*. In contrast to our other activities, this project requires running a public service and supporting its secure operation on a daily basis, while we scale up the number of servers and users.

We will explore the possibilities to install a supervisory board that oversees the management board. This is an important step to achieve oversight and stronger accountability and a prerequisite to allow compensation for management board members.

In our operation, increasing our R&D capacity is the most important objective. Our maintainers are the driving force of our impact; creating stable work environments and facilitating them to do their best work is the way forward.

Thanks to our supporters and funders

nlnet FOUNDATION

Canonical

ISRG

ASTRAL

Chainguard

Ministerie van Binnenlandse Zaken en Koninkrijksrelaties

aws

Sovereign Tech Agency

ICANN

MEINBERG

tweede golf

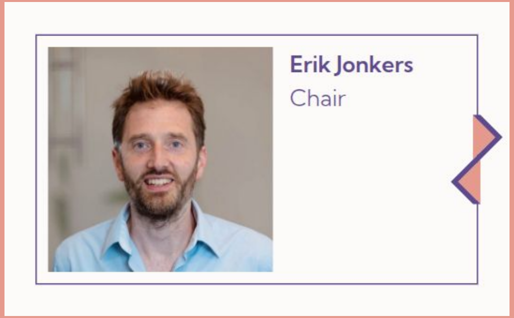
NGI ZERO CORE

Getting in touch

Trifecta Tech Foundation relies entirely on donations and sponsorships to maintain our projects and start new developments.

Your support backs our mission to strengthen critical infrastructure software.

Interested in supporting us?
Contact Erik Jonkers, erik@trifectatech.org.



ANBI information & colophon

Statutory name:	Stichting Trifecta Tech Foundation
Statutory seat:	Nijmegen, The Netherlands
Legal form:	Foundation (Dutch: Stichting)
Date of incorporation:	April 17, 2024
ANBI information:	Available via the official ANBI register[ANBI]; search on RSIN number

Address:	Castellastraat 26, 6512 EX Nijmegen, The Netherlands
Fiscal registration (RSIN)	866479004
Chamber of Commerce no:	93645546 (Dutch: Kamer van Koophandel)
Contact email:	contact@trifectatech.org
Contact phone:	+31243010484

Bylaws:	Bylaws in English and Dutch, see About page[about].
Annual reports:	Annual report, financial statement & standard form, see Report page[report].

Board	**Role**	**Start of term**
Erik Jonkers	Chair	April 17, 2024
Hugo van de Pol	Secretary	April 17, 2024
Marlon Baeten	Treasurer	April 17, 2024

References

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[report] https://trifectatech.org/reports/ < back to text